

Harsh Gupta

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EDUCATION

Indian Institute of Technology Madras

Bachelor of Science, Data Science and Applications, CGPA 8.7

Chennai, Tamil Nadu

Sept 2025

WORK EXPERIENCE

Gabeo.AI (California, USA)

Lead - Machine Learning Engineer

WFH

November 2024 – Present

- **Architected and deployed ZOE**, an AI-powered multi-agent claim denial recovery platform using **Google ADK, Vertex AI (Gemini 2.5 Pro/Flash), BigQuery, and Pinecone**; orchestrated 7 specialist agents (claim analysis, historical pattern matching, policy research, contract analysis, cross-validation, research enhancement, and appeal generation) achieving **\$200K+ in recovered revenue for a single healthcare client** and \$100K+ for another, with multiple new clients onboarding.
- Designed a **self-learning Knowledge Brain** backed by BigQuery that recalls successful denial resolution strategies and stores new patterns post-analysis, enabling the system to improve autonomously; implemented **CARC/RARC code classification**, EDI 835/837 parsing, and insurance-type-aware appeal generation using the 3C framework (Complaint, Cause, Correction).
- Built a **real-time WebSocket pipeline** (Flask-SocketIO) with thread-safe session management and cancellation tokens for concurrent multi-user claim processing; integrated **OCR (Docling, PyPDF2, Gemini vision)** for scanned medical records, GCS-based contract upload via Gemini File API, and **OpenTelemetry tracing** for production observability.
- **Designed and built ARIA**, a healthcare claims adjudication engine that determines financial responsibility (IPA/Plan/Hospital) against DOFR contracts using **Pinecone vector search, FlashRank cross-encoder reranking, and structured LLM output with ground-truth validation**; flagged **\$600K+ in overpay/underpay discrepancies** for a major health system, driving significant cost recovery.
- Engineered a **multi-stage cost-optimized inference pipeline** combining signature caching, deterministic rule-based resolution, batch API processing, and embedding LRU caching; built a **dual LLM provider abstraction** (Gemini/Claude) with prompt caching and structured JSON output schemas to eliminate hallucination and reduce per-claim processing cost by up to **70%**.
- Implemented **domain-specific code classification** across 100+ healthcare service categories, billing context validation, out-of-area provider rule enforcement, and a **comprehensive audit trail** with step-level decision tracing for regulatory compliance.

Sudha Gopalakrishnan Brain Centre (IIT Madras)

Machine Learning Research Intern

Chennai, Tamil Nadu

August 2024 – December 2025

- Trained SOTA models (YOLOv8, U-Net) for brain image segmentation to segment various parts of the brain at different hierarchical levels.
- Worked on Text2SQL RAG system for question answering about brain data and the Neuro Voyager Platform.

Metagauss (Toronto, Canada)

Data Science Intern

WFH

December 2023 – April 2024

- Developed a Retrieval-Augmented Generation (RAG) system to address inquiries from recently onboarded employees regarding hiring process documentation and terms and conditions.
- Utilized various technologies and frameworks including ChromaDB, ChatGPT API, and AWS in project development.

SKILLS

- **Technical:** Python, SQL, BASH, Data Structures
- **Frameworks:** Flask, Flask-SocketIO, FastAPI, Vue, Bootstrap, Jinja, Google ADK
- **Libraries:** Scikit-learn, NumPy, Pandas, PyTorch, Hugging Face, LangChain, LangGraph, CrewAI, SpaCy, OpenTelemetry, FlashRank
- **AI/ML:** Gemini (Vertex AI), Claude (Anthropic), Prompt Engineering, RAG, Multi-Agent Systems, Fine-Tuning (BERT, Qwen, Gemma)
- **Cloud & Data:** GCP (BigQuery, Cloud Run, GCS, Secret Manager), Pinecone, ChromaDB, SQLite, MySQL, Postgres, Neo4j, DuckDB

PROJECTS

Scalable Heart Disease Prediction API with Kubernetes & Trustworthy AI ([github](#))

- Engineered a production-ready FastAPI service containerized with Docker and deployed on Kubernetes, integrating automated MLOps pipelines for drift detection, SHAP explainability, and fairness auditing.

Explainable Stacking Ensemble for Financial Fraud Detection ([github](#))

- Developed a stacking ensemble (XGBoost, LightGBM) for financial fraud detection, achieving 99.6% recall and 99.9% precision using SHAP-based explainability and custom threshold optimization.